Machine Learning in Engineering Science
Week 9: Find, Explain, Change, Preserve

NCKU + LAiMM / Prof. Chi-Hua Yu

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1 Week 9: find, explain, change, preserve

Monday, 2 November 2026, 14:00–17:00. Use one complete Exercise notebook throughout the three one-hour modules. H1 has 45 minutes of explanation and 15 of practice; H2 has 15 and 45; H3 has 60 minutes of practice. Eight core tasks take 100 minutes, leaving 20 practice minutes for peer discussion, export and optional work. This is in-class formative practice; designated Homework and written/oral examinations are unchanged.

The goal is to explain where a predictor fails, connect a visible error pattern to a missing relationship, separate two kinds of change and preserve evidence with an honest replay scope. Prerequisites are held-out evaluation, a line and a square, and Python arrays. The core lesson requires no calculus, GPU, paid AI tool, API key or external dataset.

1.1 The four questions

1. Which cases are weak, and how many are there?
2. Which missing relationship explains the diagnostic pattern in this constructed example?
3. Did the input cases change, or did the target relationship change?
4. Which parts of the saved result can another person actually check?

A successful execution is evidence about this code and these observations. It does not establish validity for a physical system or deployment. Although these checks also matter for LSTM and Transformer predictions, we use transparent synthetic regression to make the mechanism visible.

1.2 Use the notebook independently

Save a copy, run Setup once and complete Tasks 1–8 in order. Each task provides Markdown instructions, a prediction cell, executable scaffold, stopping condition and explanation cell. The worked notebook is an alternative with example responses, not another prerequisite. If the runtime resets, run the same file from the top. Save the edited notebook and download the ZIP before the temporary runtime ends.

2 Task 1: five errors before a model library

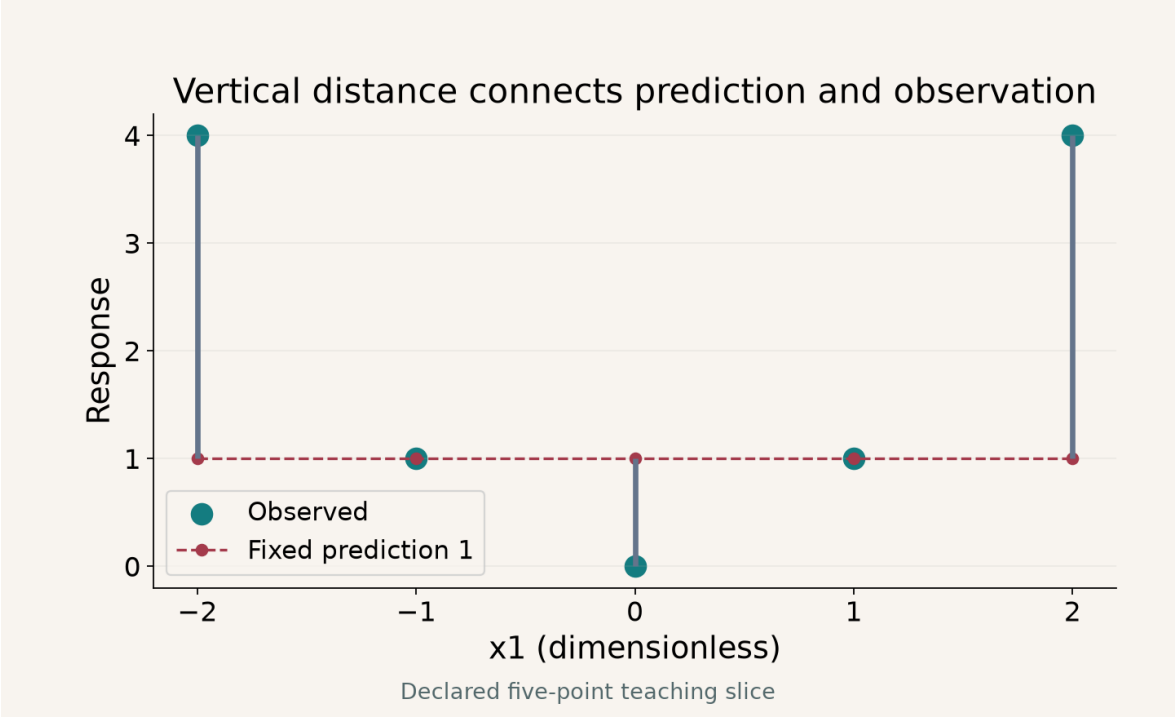
We declare the residual sign $e_i = \widehat{y}_i - y_i$: positive means too high and negative means too low. Hold $x_0 = 0$ and noise at zero. The known synthetic response is x_1^2. A deliberately fixed horizontal prediction of 1 gives:

x_1	Observation	Prediction	Residual	Absolute error
-2	4	1	-3	3
-1	1	1	0	0
0	0	1	1	1
1	1	1	0	0
2	4	1	-3	3

This horizontal prediction was not fitted to these five points and is not claimed optimal for them. It is an explicit teaching approximation, separate from the pipelines trained in Task 3.

$$\sum_i |e_i| = 7, \quad \sum_i e_i^2 = 19, \quad \text{MAE} = 7/5 = 1.4, \quad \text{RMSE} = \sqrt{19/5} \approx 1.949.$$

Both MAE and RMSE have response units; MSE has squared response units. Squaring makes a large error contribute disproportionately before the final square root. Neither metric is universally best: state the task and error costs. For the same equally weighted rows, RMSE is at least MAE.



Noise-free five-point teaching slice. Vertical segments connect the declared prediction with the observed square.

Predict and explain. Which rows are underpredicted? At $x_1 = -2$, explain why $1 - 4 = -3$ is a low prediction. Show the intermediate sums rather than calling a metric function only.

3 Task 2: the mixture can change the score

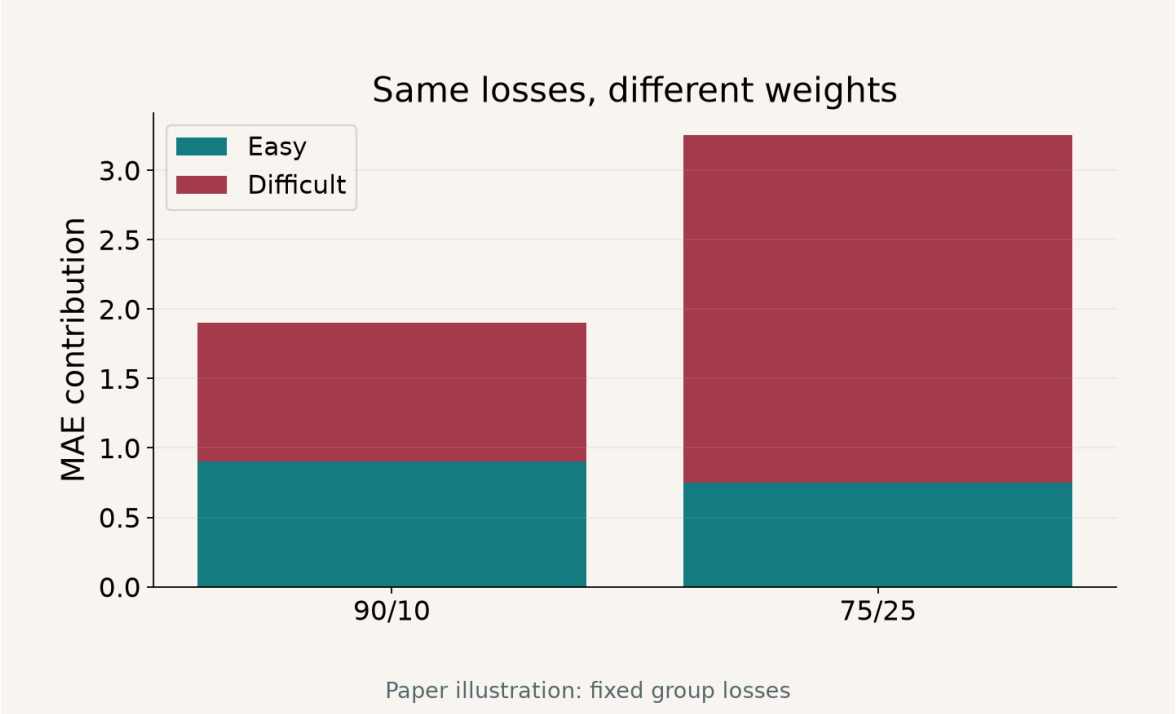
This paper illustration is separate from the 75-row test set used later. Suppose 90 easy cases each have absolute error 1 and 10 difficult cases each have error 10. Then

$$\text{MAE} = \frac{90 \times 1 + 10 \times 10}{100} = 0.9 + 1.0 = 1.9.$$

Change only the counts to 75 easy and 25 difficult cases, keeping 100 total and both group losses fixed:

$$\text{MAE} = \frac{75 \times 1 + 25 \times 10}{100} = 0.75 + 2.5 = 3.25.$$

The changed aggregate does not require changed within-group errors. More generally, for group fraction q and these two fixed losses, $\text{MAE}(q) = (1 - q)1 + q10 = 1 + 9q$. In real observations, a changed aggregate alone does not identify which mechanism changed.



Stacked contributions separate each group loss from its prevalence. The interactive slide changes the difficult-case count only.

Stopping condition. Explain which two quantities stayed fixed and which changed. Do not report 90/10 or 75/25 as the actual fitted-model test counts.

4 Task 3: one transparent main experiment

The notebook generates $n = 500$ independent synthetic rows. Each row contains dimensionless inputs x_0, x_1, x_2 drawn independently from a standard normal distribution. Noise is independent with standard deviation 0.2:

$$y = 2x_0 + x_1^2 + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, 0.2^2).$$

x_0 contributes linearly, x_1 contributes through its square and x_2 is a negative control unused by the generator. The response has synthetic response units. These are neither measured equipment data nor financial forecasts. Knowing the constructed generator permits a representation explanation without claiming physical discovery.

Before fitting, seed 2026 fixes the generated sample and first split. A 70/30 split yields 350 training IDs and 150 held IDs; seed 2027 divides held IDs equally into 75 validation and 75 test IDs. Sort IDs within splits, assert disjointness and save exact CSV bytes with row ID and membership.

4.1 Declare the candidates before inspecting test outcomes

1. Linear: raw three inputs, training-fitted standardization, Ridge with $\alpha = 1$ and SVD solver.
2. Quadratic: append the deterministic feature x_1^2, then the same training-fitted standardization and Ridge configuration.

Centering/scaling is affine and cannot create a missing square. Adding the square is a change of feature basis. Both scalers and regression coefficients fit only the 350 training rows. Choose the candidate with smallest validation MAE; exact ties favor linear. Save that choice before displaying test metrics. The predeclared two-model test comparison diagnoses errors but does not reopen selection.

4.2 A measured reference, not a score to copy

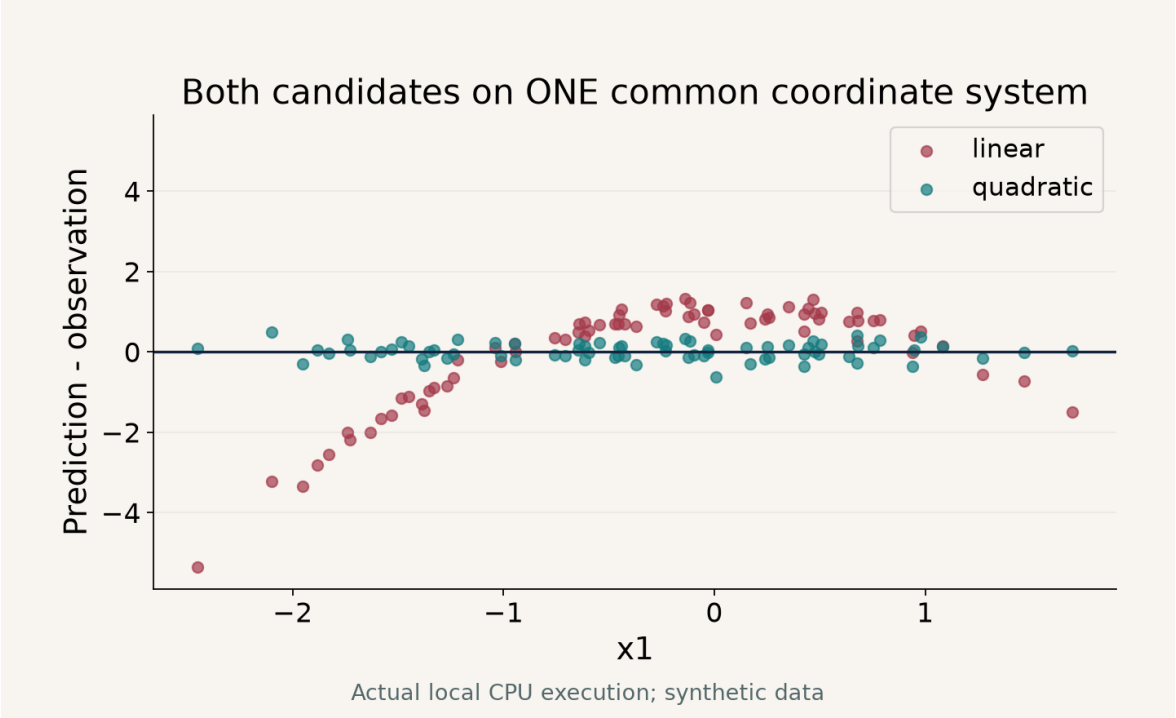
The fresh local CPU run on 27 October 2026 produced the following rounded values. Package versions are exported; tiny differences across environments are possible.

Candidate	Validation MAE	Test MAE	Test RMSE
linear	1.017086	1.031530	1.319474
quadratic	0.158188	0.164319	0.205100

The quadratic candidate was selected on validation. Do not copy these values into an unexecuted response. State the split, loss, data identity and selection rule alongside any claim.

5 Task 4: make a fair visual comparison

Plot prediction minus observation against x_1 for both candidates on the exact same 75 test rows. Use a shared horizontal range and a shared vertical range. Compare magnitude before opening the separately labeled quadratic zoom. Automatic axis scaling can make unequal errors fill two panels equally; a zoom answers a different question.



Both candidates occupy a single shared coordinate system. The linear residual leaves a visible curved pattern; the square-feature candidate has much smaller residuals.

The residual curve suggests a missing relationship. Here the square is known by construction. A curved observational residual alone does not prove physical causation.

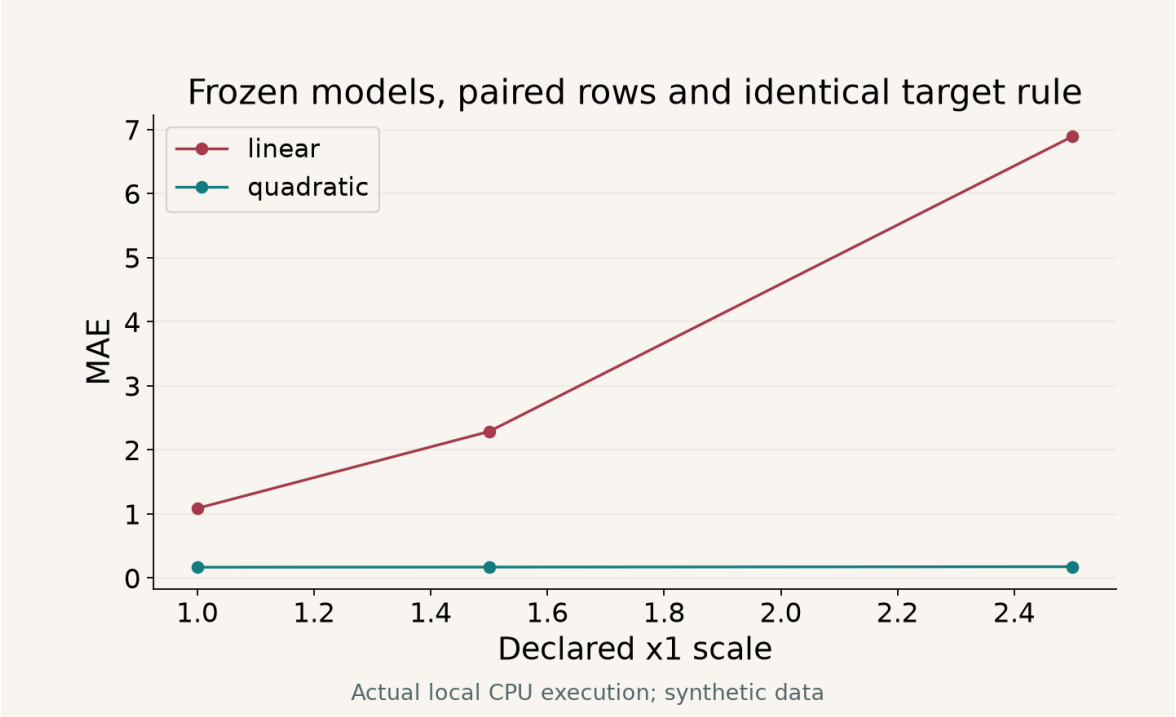
Keep the predeclared groups $|x_1| \leq 1$ and $|x_1| > 1$ fixed. Diagnose the linear candidate first, then the quadratic candidate on the same slices. Each candidate has 50 inside and 25 outside test rows. In this run, inside/outside MAE is approximately 0.775/1.545 for linear and 0.174/0.144 for quadratic. The quadratic outside group is not worse in this sample: do not force a universal claim about extreme cases.

Stopping condition. Show actual counts and group losses for both models; counts sum to 75. Do not optimize a new boundary on final test labels. Exploratory findings need a new confirmation set before renewed performance claims.

6 Task 5: input change with an unchanged rule

Use a separate seed 2126 to generate 150 paired base rows and one noise realization. Predeclare x_1 scale factors 1, 1.5 and 2.5. At each scale, change only x_1, hold x_0, x_2, row pairing and noise fixed, and recompute outcomes with the original rule $y = 2x_0 + x_1^2 + \epsilon$. Predict with both frozen candidates without refitting.

Why recompute labels? The original relationship must remain fixed. Changing a feature encoding while keeping old labels would test a different problem. Why pair rows? It makes the controlled factor explicit; it does not establish physical realism.



Actual labeled MAE at the three declared input scales. The two candidates are frozen; no training occurs during the sweep.

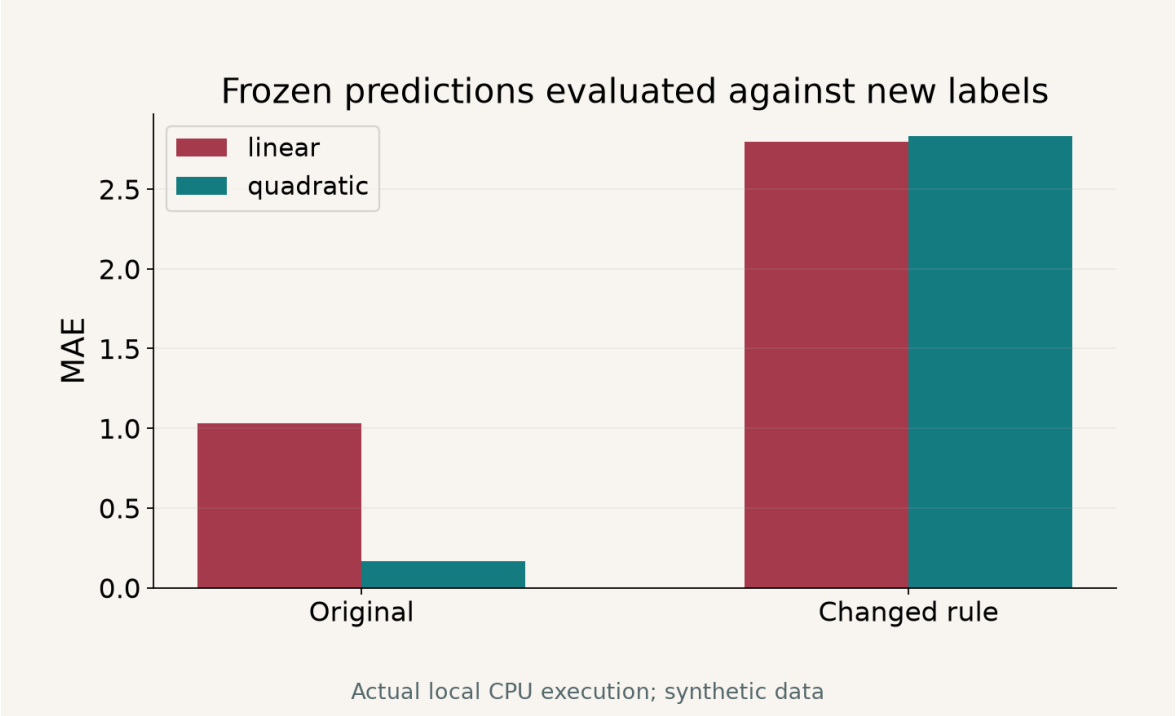
An input histogram establishes a changed input distribution in this construction; it is not an accuracy metric. Accuracy needs outcomes and a declared loss. The effect of input shift depends on the task and fitted function, so do not claim all shifts must harm all models. Snapshot recorded scaler and regression parameters before the sweep and assert exact equality afterward. This establishes unchanged recorded parameters, not arbitrary implementation state.

7 Task 6: unchanged inputs, changed target relationship

Keep the exact test inputs, row order, noise realization and existing predictions. Replace only the target rule:

$$y_{\text{new}} = -2x_0 + x_1^2 + \epsilon.$$

The input bytes and input histogram are identical by construction. The predictor also stays fixed. Only the outcomes against which it is evaluated change. Hash the input array before/after, assert identical predictions and recompute labeled errors.



The original and changed-rule errors use the same inputs and frozen predictions. Only target outcomes differ.

This constructed conditional change cannot be detected by monitoring inputs alone. Outcomes reveal its effect. It does not prove every possible relationship change must worsen every metric. In real systems, outcomes may arrive late or selectively; recording input drift and recording labeled performance are separate obligations.

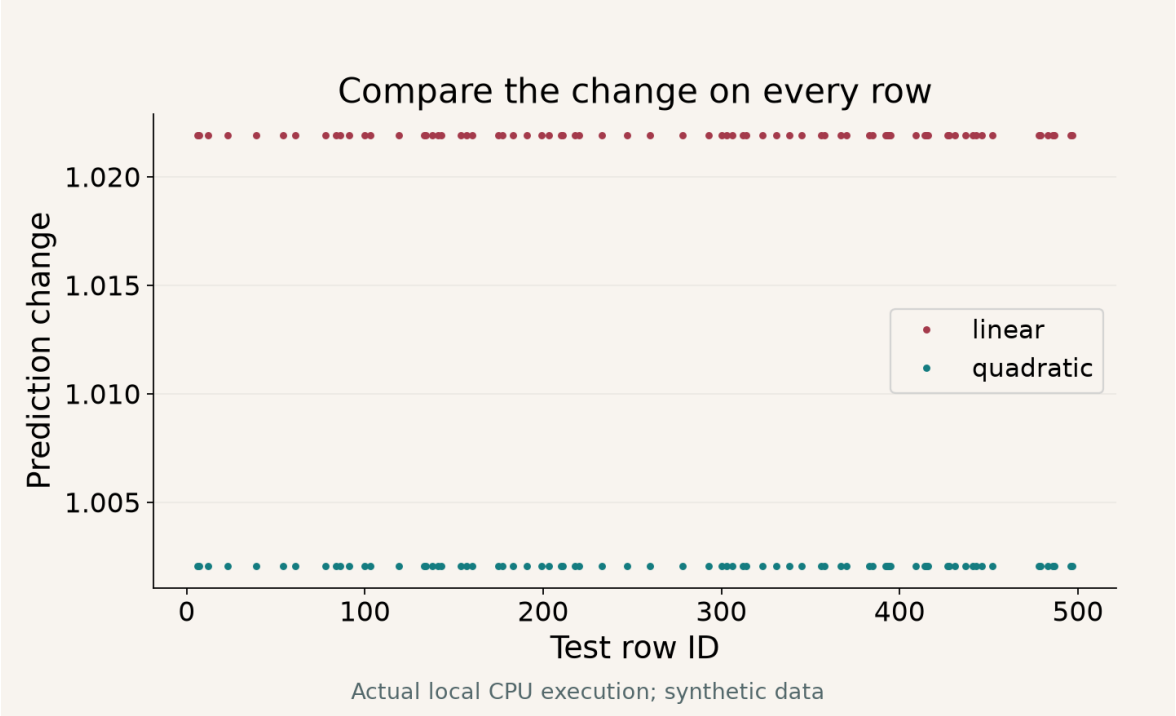
Explain the distinction. Task 5 changes cases while preserving the rule; Task 6 preserves cases while changing the rule. Both keep model parameters fixed. Identify the matching evidence rather than naming both situations only as "drift."

8 Task 7: check an explanation of the fitted function

Increase original-coordinate x_0 by 0.5 while holding all other inputs fixed. Standardization uses $z_j = (x_j - \mu_j)/s_j$. If the standardized Ridge coefficient is θ_0, the raw-coordinate slope is $\beta_0 = \theta_0/s_0$. In these additive models, no square or interaction contains x_0, so every row has

$$f(x + 0.5e_0) - f(x) = 0.5\beta_0.$$

Predict this difference before calling the model. Compare the entire 75-row difference vector against the formula, not just its mean. The numerical check uses $|a - b| \le 10^{-10} + 10^{-10}|b|$.



Observed prediction changes for every test row when only x0 increases by 0.5. Each model has its own fitted raw-coordinate slope.

No new target labels enter this check. It verifies a property of the fitted mathematical function and the coordinate convention. It is neither a physical intervention nor a new accuracy estimate. A standardized coefficient refers to one standardized feature step; do not apply it directly to a raw-input step.

9 Task 8: CSV readback and an honest handover

Read the exact frozen CSV bytes and compare their SHA256 with the saved reference. Parse and sort by stable row ID; assert 500 unique IDs and the original split membership. Clone the already-selected candidate, fit only CSV training rows and compare the full, aligned 75-row test prediction vector with the reference. This run produced maximum absolute difference zero, under the declared tolerance.

Scope: same-kernel CSV readback consistency. Model definitions, configuration, selected candidate and installed dependencies still come from the current kernel. This is not independent-process or fresh-environment reproduction. Week 10 extends the execution contract. A checksum means matching bytes against a reference; it does not authenticate the sender or establish scientific correctness.

9.1 Export and explain-back

Complete 19 response fields: eight predictions, eight explanations, controlled change, limitation and AI-use disclosure. Empty fields produce a draft. Nonempty fields mean ready for peer discussion, not automatically correct. The ZIP contains frozen CSV, row-aligned predictions, metrics, configuration, versions, figures, responses, supplied source snapshot and hash manifest. Save the edited notebook separately if you changed code.

A partner should select one actual result and ask: what changed, what stayed fixed, what supports your explanation and what remains untested? Locate the matching exported record. A ZIP that opens successfully does not itself establish a correct model or conclusion.

9.2 Troubleshooting

A missing variable usually means a skipped cell or runtime reset: run from Setup in numeric order. Keep assertion failures rather than copying a reference metric. Compare data identity, row alignment, configuration and package versions before attributing a difference to the algorithm. Download the ZIP through the Colab Files sidebar or Jupyter file link before leaving the temporary runtime.

9.3 Summary

Find weak cases before trusting an average. Explain the missing shape using a checked mechanism. Separate input changes from target-rule changes. Test an explanation against the fitted function. Preserve evidence and name the replay boundary.

10 Supplementary reasoning: optional, not extra core tasks

10.1 Conditional risk and weighted groups

For a fixed fitted function f, declared loss L and joint distribution P, population risk is $R_P(f) = \mathbb{E}_P[L(Y, f(X))]$. A finite test mean estimates this quantity only under sampling assumptions. A result on P does not automatically transfer to Q. For exhaustive groups, population risk is a prevalence-weighted sum of conditional risks; finite group MAE uses the same identity exactly. Group RMSE must be pooled through squared errors:

$$\mathrm{RMSE}_{\mathrm{all}} = \sqrt{\frac{\sum_g n_g \mathrm{RMSE}_g^2}{\sum_g n_g}},$$

not a simple mean of group RMSE values.

10.2 Why the population linear approximation misses the square

For independent standard-normal coordinates and mean-zero noise, the least-squares projection of $2x_0 + x_1^2 + \epsilon$ onto an intercept and linear inputs is $1 + 2x_0$: $\mathbb{E}[x_1^2] = 1$, $\mathbb{E}[x_1^3] = 0$ and independent cross-covariances vanish. Its residual is $1 - x_1^2 - \epsilon$. This explains the curved pattern. The finite, regularized Ridge fit is not exactly this population projection; measured coefficients and residuals need not equal the ideal formula. The Task 1 fixed line is a noise-free visual slice of this motivation, not a fit to its five points.

Normal distributions have unbounded support. Scaling x_1 changes its distribution and the frequency of tail cases, not its theoretical support from finite to infinite. A finite training sample still leaves sparsely observed regions. Describe coverage honestly rather than claiming every wider case is outside normal support.

10.3 Recovering original-coordinate coefficients

For feature vector $\phi(x)$, standardized predictor $b_s + \sum_j \theta_j(\phi_j - \mu_j)/s_j$ has raw coefficients $\beta_j = \theta_j/s_j$ and intercept $b = b_s - \sum_j \theta_j \mu_j / s_j$. This coordinate conversion explains Task 7; it does not turn association into causality.

10.4 Optional A: encoding check

Multiply the dimensionless x_0 encoding by 100 while retaining original targets and frozen models; measure the error, then decode by dividing by 100 and compare aligned predictions. This tests a representation mismatch. It is not a physical unit conversion and differs from Task 5, which deliberately recomputes outcomes to preserve the target rule.

10.5 Optional B: fixed-model bootstrap

Within each fixed slice, sample absolute errors with replacement 300 times and report percentile bounds for their mean. This assumes IID cases within the slice and conditions on the already

fitted function. It excludes training-set variation, arbitrary future distribution change and many measurement errors. A narrow interval does not remove those limits. Optional results remain separate from the eight-task completion count.

11 Primary references

- Monitoring ML pipelines: https://developers.google.com/machine-learning/crash-course/production-ml-systems/monitoring.
- Leakage-safe pipelines: https://scikit-learn.org/stable/common_pitfalls.html.
- Regression metrics: https://scikit-learn.org/stable/modules/model_evaluation.html#regression-metrics.

References checked 27 October 2026. All examples and data are synthetic teaching fixtures.